HeatLens: Visual Analytics for Urban Heat Risk Assessment and Mitigation Planning

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1 Introduction & Motivation

Extreme heat is the deadliest weather hazard in the United States. However, existing tools for analyzing heat risk rely on static maps and do not integrate social vulnerability, support interactive exploration, or enable testing of mitigation strategies. Prior work on urban heat islands [1] identifies spatial patterns but rarely links them to health outcomes or actionable decisions.

Our Goal: We propose **HeatLens**, an interactive visual analytics system for identifying high-risk regions, understanding drivers of heat risk, and evaluating mitigation strategies. The system targets city planners, public-health agencies, and emergency managers. Because extreme heat disproportionately affects older and low-income populations [2], interpretable heat-risk analysis is critical. We use **California** as a case study due to its diverse climates, severe heat events, and detailed public-health data.

2 Data

We use three free datasets covering recent years:

- **NOAA GHCN-Daily**: daily temperature from about 1,500 California weather stations, yielding $\sim$3 million daily records over our study period
- **Tracking California**: county-level heat-related ED visit rates (the target variable's native granularity), used as ground truth for our risk model
- **US Census ACS**: vulnerability features like age, income, race, and AC ownership at the county level, plus tract level ($\sim$9,000 tracts with dozens of demographic variables) for spatial analysis

Combining these is not trivial—we need to derive county-year climate features from millions of daily station records, match weather stations to county polygons, and handle missing data in rural areas.

3 Related Work and Limits of Current Practice

Our work builds on three research threads. (1) *Time-series anomaly detection*: Matrix Profile [3] and AER [4] support anomaly detection in temporal data, but assume consistent temporal signals rarely available in public-health datasets. (2) *Spatiotemporal visual analytics*: TPFlow [5] and urban heat island studies [1] analyze large-scale spatial patterns, but rarely connect them to health outcomes or decision-making. (3) *Human-centered visual analytics*: InclusiViz [6] and prior work on heat risk definitions [2] support interactive analysis of complex social data, but existing systems lack interpretable models and mitigation analysis.

Overall, existing heat-risk systems combine climate, demographic, and health data, but rely on predefined thresholds and limited interaction. In contrast, our approach integrates explainable ML, counterfactual SHAP for intervention-aware explanations, and interactive what-if simulation within a unified visual analytics system for California.

4 Proposed Approach and Expected Innovations

Algorithmic side. We engineer **annual climate features** from daily station data—summer-averaged maxima, heatwave-day counts, consecutive hot-day streaks, warm-night frequencies, and tail-percentile statistics—to capture the multi-faceted ways heat affects health beyond a simple summer average. We train an **XGBoost** model to predict county-level heat-related ED visit rates, with **SHAP values** explaining each prediction. Our key algorithmic contribution is **counterfactual SHAP**: when users adjust intervention sliders in the visualization, we recompute SHAP values on the modified instance and surface the change in feature contributions, so users see not just the new prediction but which features drove the change. We use leave-county-out cross-validation to handle spatial correlation in our 40-county training set, and compare against linear regression and a temperature-only threshold baseline.

Visualization side. HeatLens presents four linked views: (i) a **map overview** of county risk over time, (ii) a **climate-feature detail** view showing which features drive each county's risk, (iii) a **vulnerability breakdown** via SHAP force plot, and (iv) a **what-if simulator** where intervention sliders (e.g., increased AC coverage or tree canopy) update predictions in real time. The simulator is our most distinctive contribution—existing tools such as the CDC Heat & Health Tracker visualize current data but do not support testing interventions.

Why this will succeed. The core algorithms (XGBoost, SHAP, feature engineering) all have mature open-source libraries, and our counterfactual SHAP extension is a thin wrapper computable in a few lines. The California-scoped data fits on a laptop, and the team is split across data, ML, and front-end roles for parallel development.

5 Evaluation Plan

First, we will evaluate the predictive accuracy of the heat-risk model by comparing the XGBoost model with linear regression model and a temperature-only threshold baseline. Second, we will evaluate the interpretability of the model by using SHAP values to identify the most important climate and factors contributing to heat risk. Finally, we will evaluate the what-if simulator qualitatively and focus on whether users can understand how changes in factors such as AC access or tree canopy affect model predictions.

6 Risks, Payoffs, and Cost

Risks. The main risks are the small county-level health dataset and potential downward bias in 2020–2021 ED visit rates due to pandemic care avoidance. **Mitigations.** We will use leave-county-out cross-validation with strong regularization, treat 2020–2021 as a sensitivity analysis, and clearly label the simulator as exploratory scenario analysis. **Payoffs.** HeatLens provide a reusable visual analytics pipeline for climate-health risk assessment and could be extended to related domains such as air quality, wildfire smoke, and flooding risk.**Cost.** All data and compute are free.

7 Timeline and Checkpoints

Task	**Role**	**May 5–12**	**May 13–20**	**May 21–26**	**May 27–June 5**
Data collection	Yan Liang				
Feature engineering	Peiyu Lin				
Visualization interface	Pablo Rodriguez				
Risk model and SHAP	Peiyu Lin				
What-if simulator	Pablo, Yan Liang				
Progress report	All members			**Due May 26**	
Final report	All members				**Due June 5**

Midterm: Data collection, training model and evaluated with SHAP explanations, and finish at least one linked visualization view. **Final:** Whether HeatLens is usable as a complete system.

References

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[4] L. Wong, D. Liu, L. Berti-Equille, S. Alnegheimish, and K. Veeramachaneni, "AER: Auto-Encoder with Regression for Time Series Anomaly Detection," *IEEE Big Data*, 2022.

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[6] Y. Yu, Y. Wang, Y. Zhang, . . . , and D. Liu, "InclusiViz: Visual Analytics of Human Mobility Data for Understanding and Mitigating Urban Segregation," *IEEE TVCG*, 2025.